

From Physics-Based Earthquake Catalogs to Seismic Hazard: First Steps with VEGA

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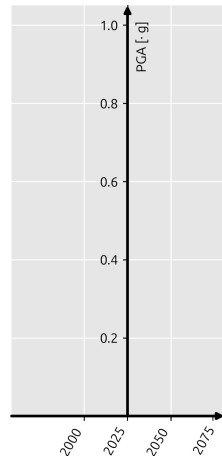
École Normale Supérieure, Paris
August 2026

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Estimates the probability that ground motion at a site exceeds a given intensity level.

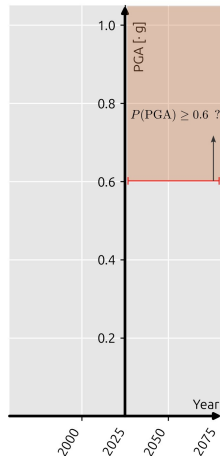
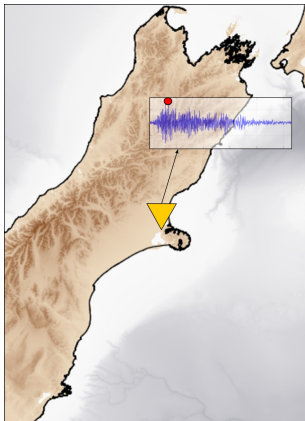
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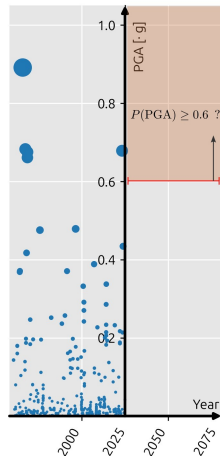
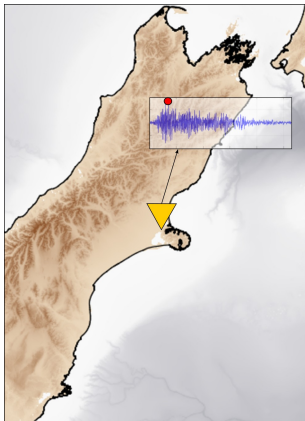
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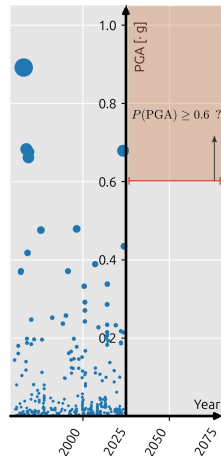
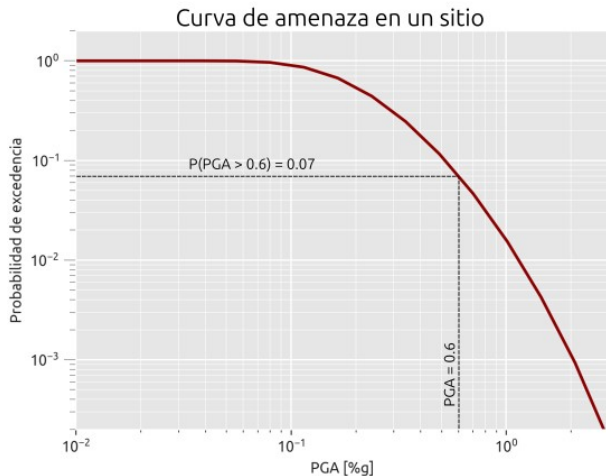
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How a PSHA model is built

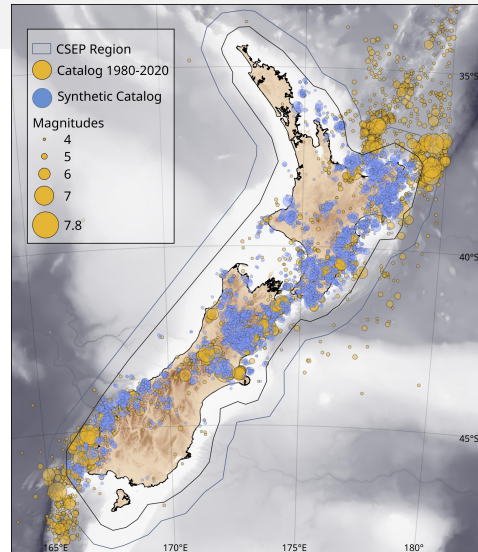
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Earthquake forecasting (probabilistic)

- Identify seismogenic sources.
- Space–time distribution of seismicity.
- Frequency–magnitude distribution.



One realization of a **stochastic catalog** model (ETAS) for New Zealand crustal seismicity (Iturrieta et al., 2024). Occurrence is statistical, not physical.

How a PSHA model is built

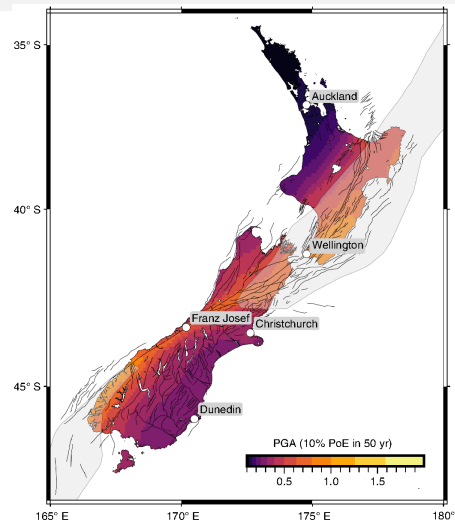
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New Zealand seismic hazard model, 2022 (Gerstenberger et al., 2023).
PGA with 10% probability of exceedance in 50 years ($V_{S30} = 250$ m/s).

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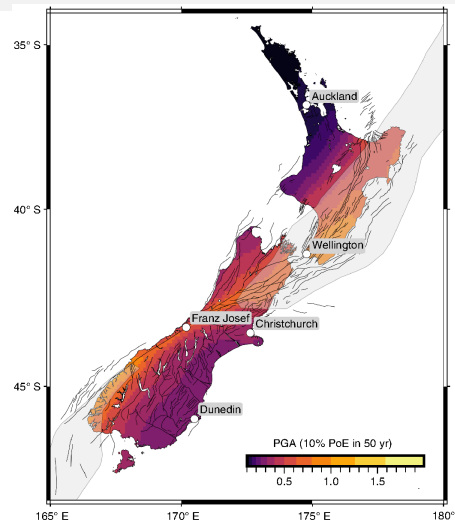
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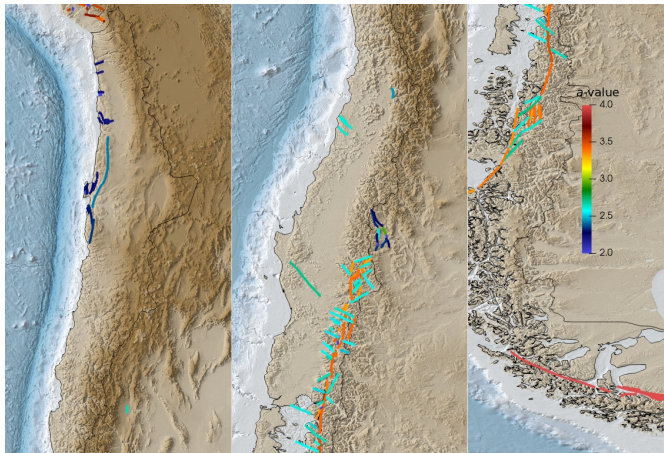
How crustal faults enter a hazard model today

- Gutenberg–Richter or characteristic FMD, *imposed* on each fault.
- M_{max} from fault geometry or scaling relations.
- Solve for the rate a from a long-term moment budget.

$$\dot{M}^{\text{long-term}} = \dot{M}^{\text{seismic}}$$

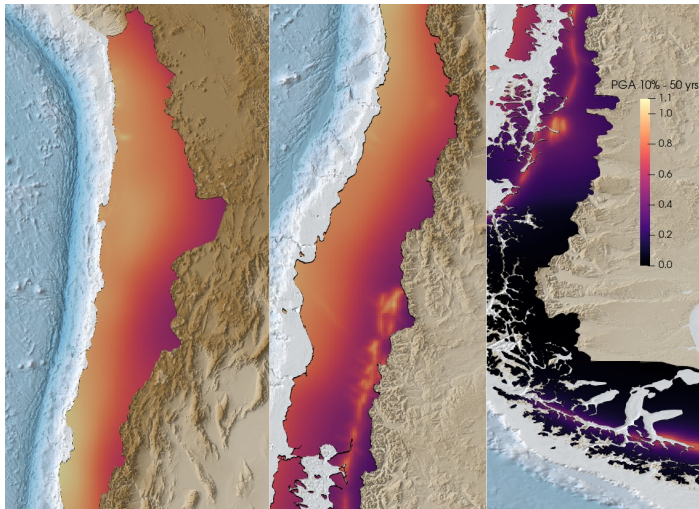
$$\dot{M}^{\text{long-term}} = \phi \mu L W \dot{s}_r$$

Where $\dot{s}_r$ is the slip rate and ϕ the seismic/aseismic coupling factor.



Fault activity rates for the Chilean model, derived from slip rates.

Seismic hazard map for Chile: faults do matter



Hazard map **including** crustal faults, subduction and intraplate earthquakes: PGA with 10% probability of exceedance in 50 years, $V_{S30} = 250$ m/s. Computed with OpenQuake (Pagani et al., 2012).

The assumptions we would like to relax

Everything in the previous slide rests on a set of convenient assumptions:

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The question of this work

Can a **physics-based** earthquake simulator produce synthetic catalogs from which hazard is computed *without* any of these assumptions, and how different are the results?

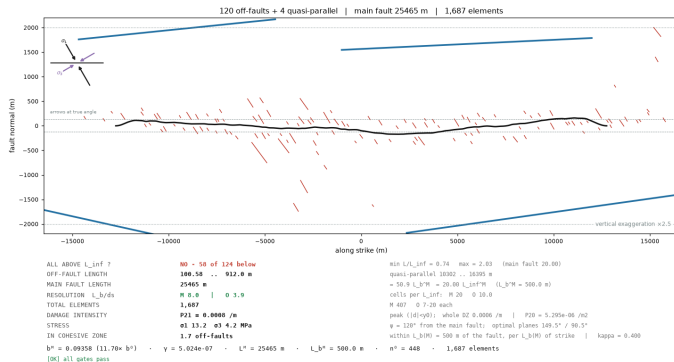
Synthetic catalogs with VEGA

What we prescribe:

- A fault network: one main fault ($L \approx 25$ km), quasi-parallel structures, and a damage zone of off-fault sources.
- Rate-and-state friction
- A background stress state and a constant shear loading rate.

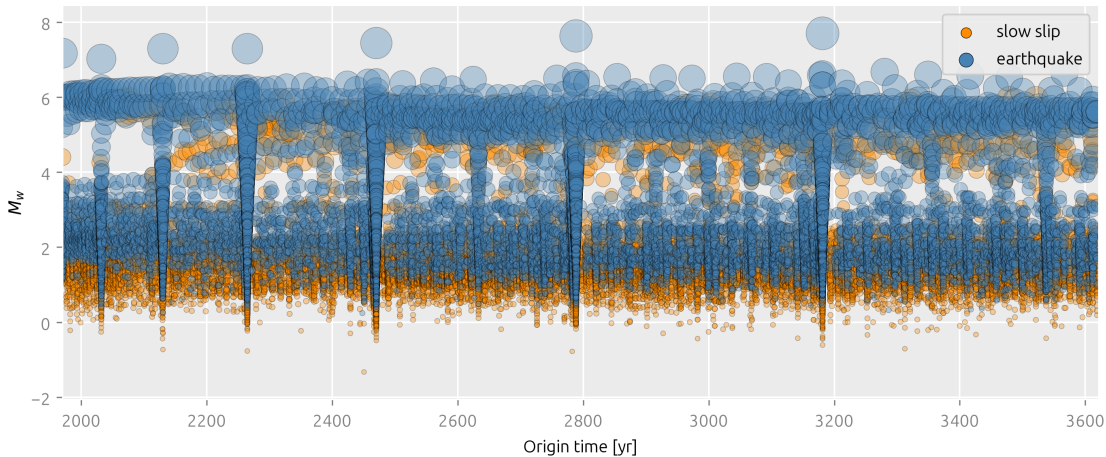
What we do not prescribe:

- When earthquakes happen.
- How big they are.
- Whether they cluster, cascade, or repeat.



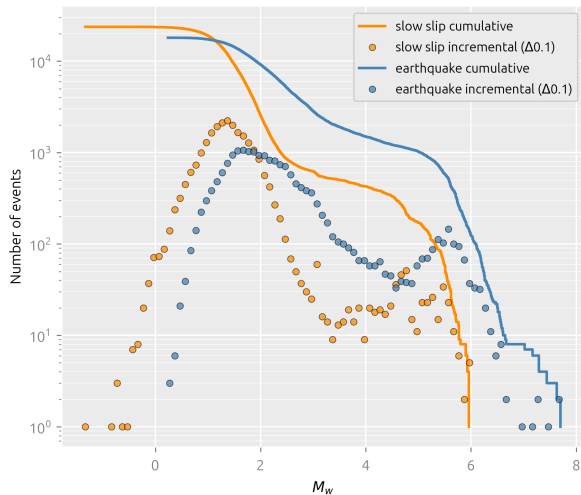
Fault network generated for the experiment.

The resulting catalog

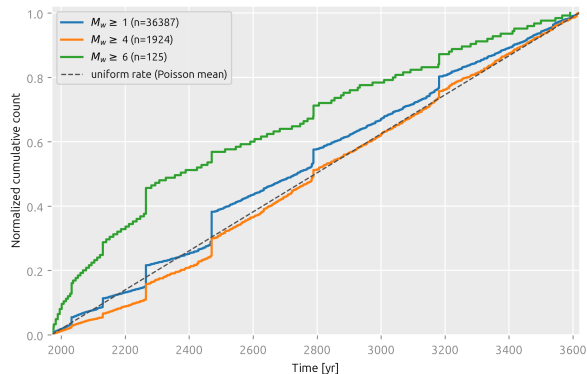


Magnitude versus time. Large events recur quasi-periodically and are followed by triggered cascades.

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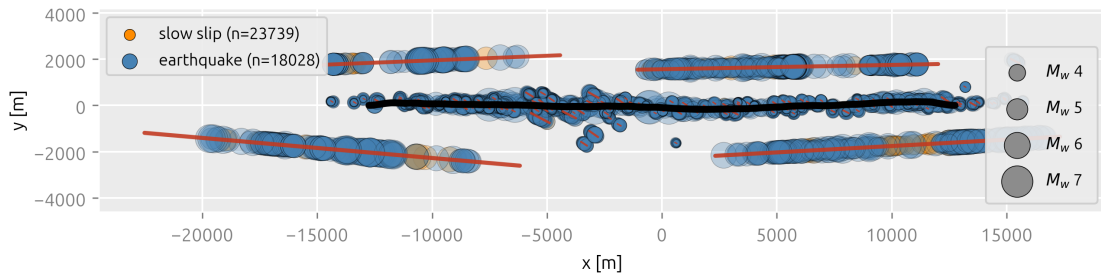


Frequency-magnitude distribution: incremental (dots) and cumulative (lines).



Normalized cumulative count against the uniform-rate (Poisson) expectation.

Spatial distribution



Epicenters of the synthetic catalog over the fault network. Marker size scales with magnitude.

From the catalog to hazard

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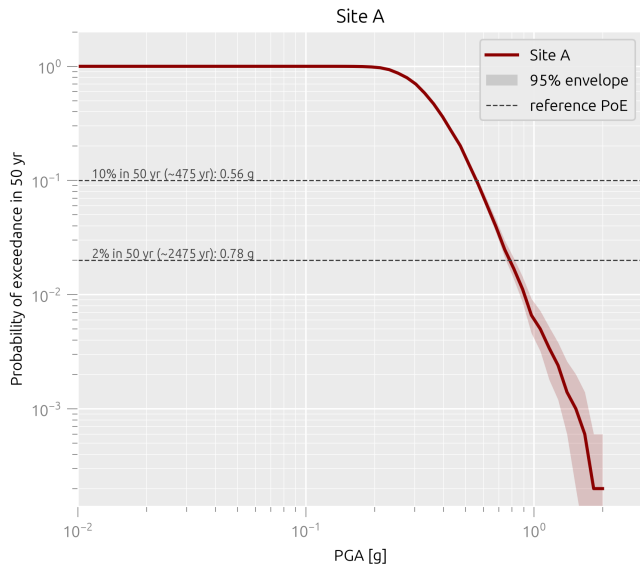
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4. Take the maximum PGA of each window.
5. The hazard curve is the empirical survival function of those maxima.

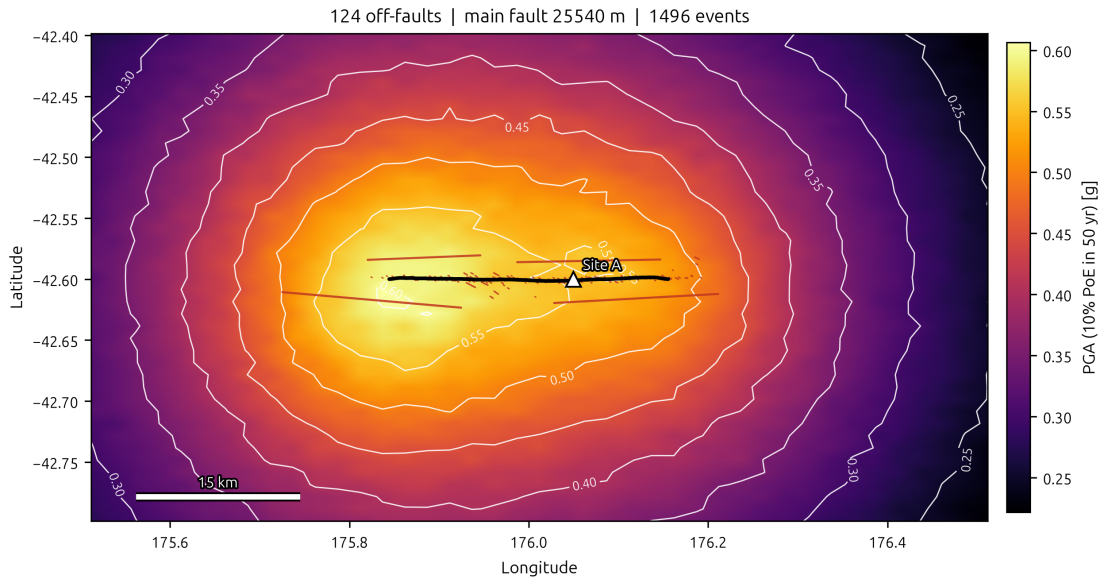
$$\hat{P}(\text{IM} > y \text{ in } T) = \frac{1}{N} \sum_{k=1}^N H(Y_{\max}^{(k)} - y)$$

where N is the number of synthetic catalogs, y is a given value of ground motion intensity (e.g., PGA) and $Y_{\max}^{(k)}$ is the maximum sampled ground motion of each synthetic catalog.

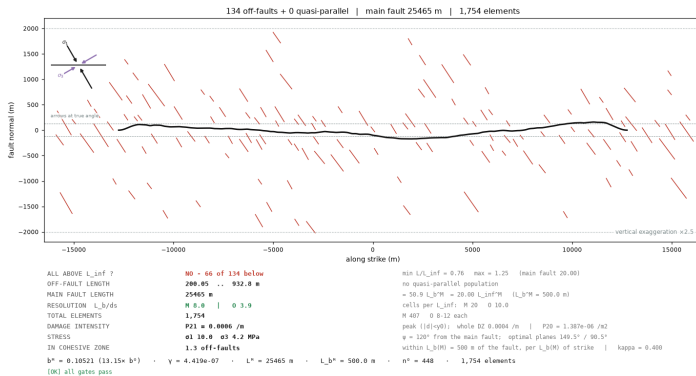
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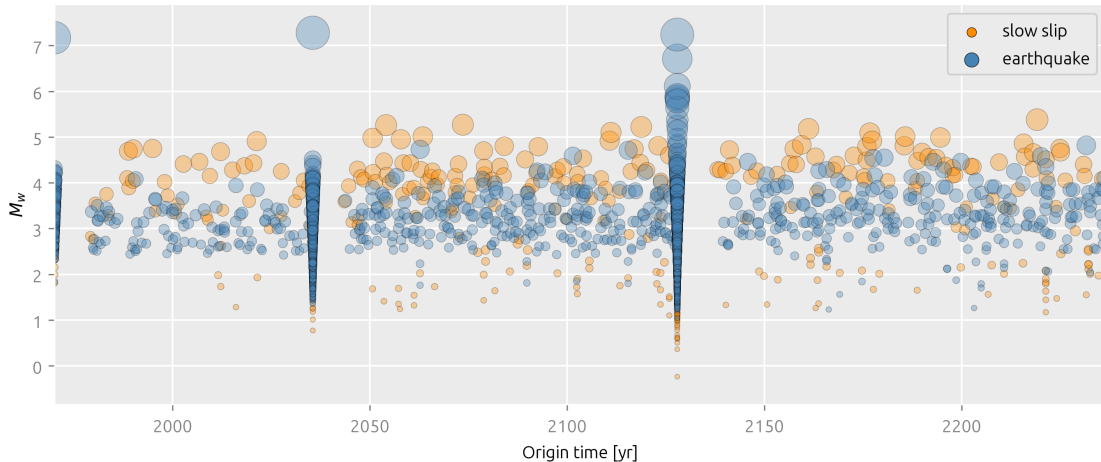


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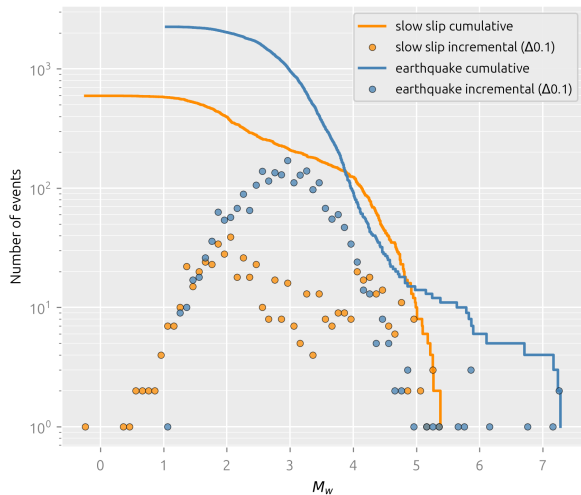
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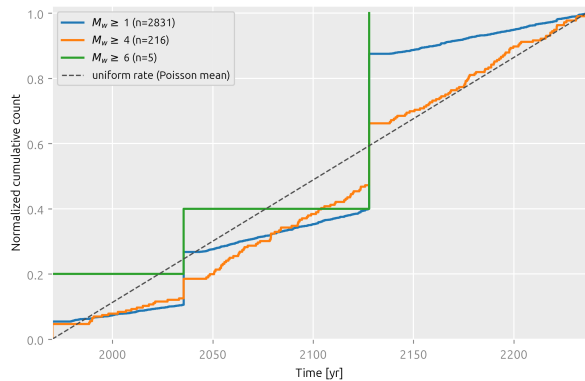


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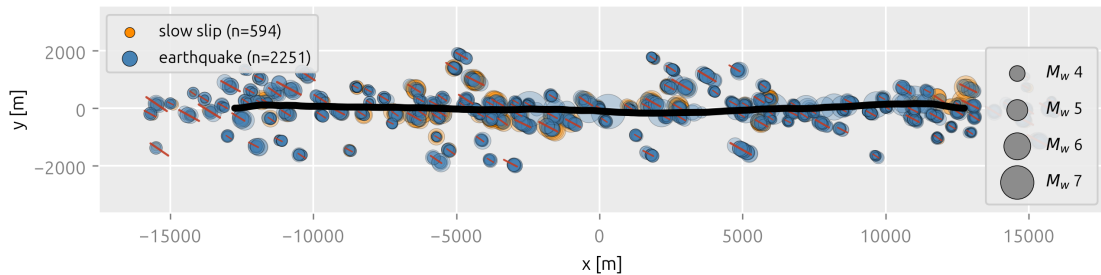


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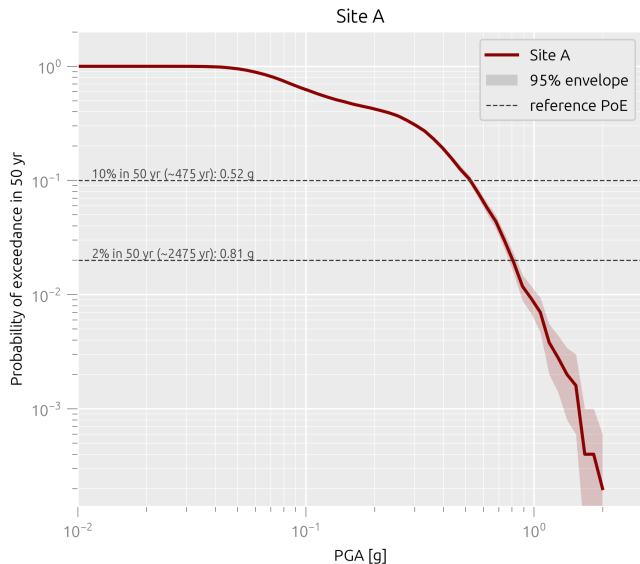
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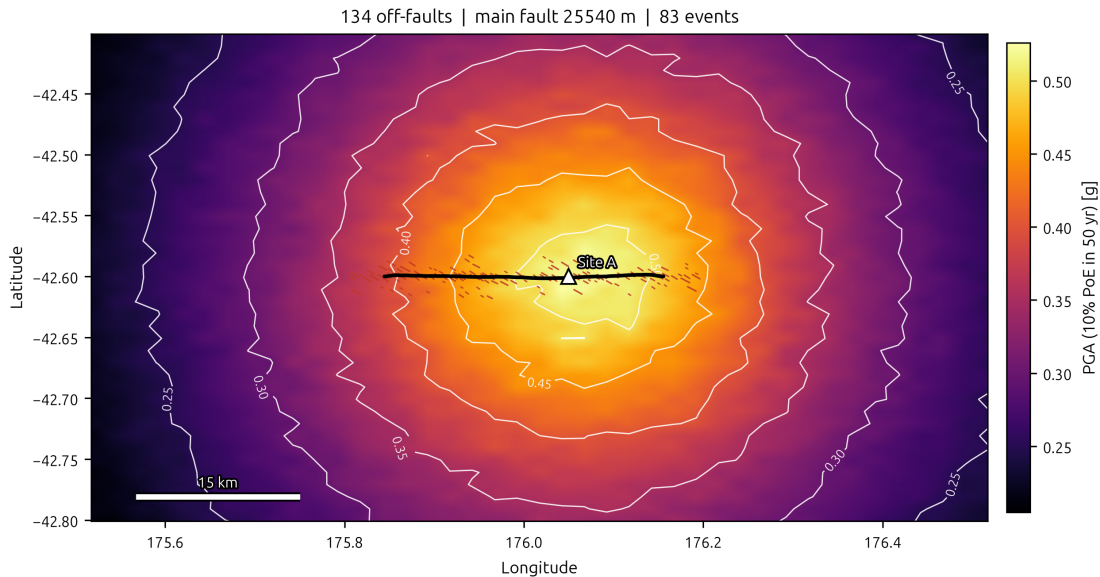


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What works:

- Geometry generator $\rightarrow$ VEGA $\rightarrow$ catalog $\rightarrow$ windows $\rightarrow$ GMM $\rightarrow$ hazard curves and maps.
- Magnitudes spanning $M_w \approx 4.5\text{--}7$ from a single network, with (almost) emergent Gutenberg–Richter behaviour, clustering and quasi-periodic large events.

Limitations learned:

- Point-source distances (finite-rupture distances missing).
- Better constrain loading rate (tested arbitrary values).
- Statistical resolution is bounded by catalog length: the tail of the curve is limited by Δ/T .
- Still a gap in the GR distribution ($M_w \in [3, 6]$)

Next: